

Assessment tool 1 co-1

HackerRank | Prepare | Java | Introduction | Java Stdin and Stdout I | Exit Full Screen View

Problem
Most HackerRank challenges require you to read input from `stdin` (standard input) and write output to `stdout` (standard output).
One popular way to read input from `stdin` is by using the `Scanner` class and specifying the Input Stream as `System.in`. For example:

```
Scanner scanner = new Scanner(System.in);
String myString = scanner.next();
int myInt = scanner.nextInt();
scanner.close();

System.out.println("myString is: " + myString);
System.out.println("myInt is: " + myInt);
```

The code above creates a `Scanner` object named `scanner` and uses it to read a `String` and an `int`. It then closes the `Scanner` object because there is no more input to read, and prints to `stdout` using `System.out.println(String)`. So, if our input is:

```
Hi 5
```

Our code will print:

```
myString is: Hi
myInt is: 5
```

Alternatively, you can use the `BufferedReader` class.

Task

Change Theme | Language: Java 15

```
1 import java.util.Scanner;
2
3 public class Solution {
4     public static void main(String[] args) {
5         Scanner scanner = new Scanner(System.in);
6
7         int a = scanner.nextInt();
8         int b = scanner.nextInt();
9         int c = scanner.nextInt();
10
11         System.out.println(a);
12         System.out.println(b);
13         System.out.println(c);
14
15         scanner.close();
16     }
17 }
18
19
20
```

Line: 20 Col: 1

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2.

HackerRank | Prepare | Java | Introduction | Java If-Else | Exit Full Screen View

Problem
In this challenge, we test your knowledge of using if-else conditional statements to automate decision-making processes. An if-else statement has the following logical flow:

IF (A = TRUE)
Then B
Else C
End IF

```
graph TD
    Start(( )) --> A{A}
    A -- TRUE --> B([B])
    A -- FALSE --> C([C])
    B --> End((( )))
    C --> End
```

Source: Wikipedia

Change Theme | Language: Java 15

```
1 import java.util.Scanner;
2 public class Solution {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int N = sc.nextInt();
6         if (N % 2 != 0) {
7             System.out.println("Weird");
8         } else if (N >= 2 && N <= 5) {
9             System.out.println("Not Weird");
10        } else if (N >= 6 && N <= 20) {
11            System.out.println("Weird");
12        } else {
13            System.out.println("Not Weird");
14        }
15        sc.close();
16    }
17 }
18
```

Line: 18 Col: 1

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34°C Mostly cloudy | Search | 11:46 28-07-2026

3.

HackerRank | Prepare | Java | Introduction | Java Stdin and Stdout II | Exit Full Screen View

Problem

In this challenge, you must read an integer, a double, and a String from stdin, then print the values according to the instructions in the Output Format section below. To make the problem a little easier, a portion of the code is provided for you in the editor.

Note: We recommend completing Java Stdin and Stdout I before attempting this challenge.

Input Format

There are three lines of input:

1. The first line contains an integer.
2. The second line contains a double.
3. The third line contains a String.

Output Format

There are three lines of output:

1. On the first line, print `String`: followed by the unaltered String read from stdin.
2. On the second line, print `Double`: followed by the unaltered double read from stdin.
3. On the third line, print `Int`: followed by the unaltered integer read from stdin.

To make the problem easier, a portion of the code is already provided in the editor.

Note: If you use the `nextLine()` method immediately following the `nextInt()` method, recall that `nextInt()` reads integer tokens; because of this, the last newline character for that line of integer input is still queued in the input buffer and the next `nextLine()` will be reading the remainder of the integer line (which is empty).

Sample Input

```
1  import java.util.Scanner;
2  public class Solution {
3      public static void main(String[] args) {
4          Scanner sc = new Scanner(System.in);
5          int i = sc.nextInt();
6          double d = sc.nextDouble();
7          sc.nextLine();
8          String s = sc.nextLine();
9          System.out.println("String: " + s);
10         System.out.println("Double: " + d);
11         System.out.println("Int: " + i);
12     }
13 }
14
15
16
```

Line: 16 Col: 1

Upload Code as File | Test against custom input | Run Code | Submit Code

34°C Mostly cloudy | Search | 11:46 28-07-2026

4.

HackerRank | Prepare | Java | Introduction | Java Output Formatting | Exit Full Screen View

Problem

Java's `System.out.printf` function can be used to print formatted output. The purpose of this exercise is to test your understanding of formatting output using `printf`.

To get you started, a portion of the solution is provided for you in the editor; you must format and print the input to complete the solution.

Input Format

Every line of input will contain a String followed by an integer.

Each String will have a maximum of 10 alphabetic characters, and each integer will be in the inclusive range from 0 to 999.

Output Format

In each line of output there should be two columns:

- The first column contains the String and is left justified using exactly 15 characters.
- The second column contains the integer, expressed in exactly 3 digits; if the original input has less than three digits, you must pad your output's leading digits with zeroes.

Sample Input

```
java 100
cpp 65
python 50
```

Sample Output

```
=====
java 100
cpp 65
python 50
```

Code Editor:

```
1  import java.util.Scanner;
2  public class Solution {
3      public static void main(String[] args) {
4          Scanner sc = new Scanner(System.in);
5          System.out.println("=====");
6          for (int i = 0; i < 3; i++) {
7              String s = sc.next();
8              int x = sc.nextInt();
9              System.out.printf("%-15s%03d\n", s, x);
10         }
11         System.out.println("=====");
12         sc.close();
13     }
14 }
15
```

Line: 15 Col: 1

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5.

HackerRank | Prepare | Java | Introduction | Java Loops I | Exit Full Screen View

Problem
Objective
In this challenge, we're going to use loops to help us do some simple math.

Task
Given an integer, N , print its first 10 multiples. Each multiple $N \times i$ (where $1 \leq i \leq 10$) should be printed on a new line in the form: $N \times i = \text{result}$.

Input Format
A single integer, N .

Constraints
• $2 \leq N \leq 20$

Output Format
Print 10 lines of output; each line i (where $1 \leq i \leq 10$) contains the **result** of $N \times i$ in the form:
 $N \times i = \text{result}$.

Sample Input
2

Sample Output
2 x 1 = 2
2 x 2 = 4
2 x 3 = 6
2 x 4 = 8
2 x 5 = 10
2 x 6 = 12
2 x 7 = 14
2 x 8 = 16
2 x 9 = 18
2 x 10 = 20

Change Theme | Language | Java 15

```
1 import java.util.Scanner;
2 public class Solution {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int N = sc.nextInt();
6         for (int i = 1; i <= 10; i++) {
7             System.out.println(N + " x " + i + " = " + (N * i));
8         }
9     }
10 }
11
12
```

Line: 12 Col: 1

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6.

HackerRank | Prepare | Java | Introduction | Java Loops II | Exit Full Screen View

Problem
We use the integers a , b , and n to create the following series:
 $(a + 2^0 \cdot b), (a + 2^0 \cdot b + 2^1 \cdot b), \dots, (a + 2^0 \cdot b + 2^1 \cdot b + \dots + 2^{n-1} \cdot b)$

You are given q queries in the form of a , b , and n . For each query, print the series corresponding to the given a , b , and n values as a single line of n space-separated integers.

Input Format
The first line contains an integer, q , denoting the number of queries.
Each line i of the q subsequent lines contains three space-separated integers describing the respective a_i , b_i , and n_i values for that query.

Constraints
• $0 \leq q \leq 500$
• $0 \leq a, b \leq 50$
• $1 \leq n \leq 15$

Output Format
For each query, print the corresponding series on a new line. Each series must be printed in order as a single line of n space-separated integers.

Sample Input
2
0 2 10
6 3 5

Change Theme | Language | Java 15

```
1 import java.util.Scanner;
2 public class Solution {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int q = sc.nextInt();
6         while (q-- > 0) {
7             int a = sc.nextInt();
8             int b = sc.nextInt();
9             int n = sc.nextInt();
10            int sum = a;
11            for (int i = 0; i < n; i++) {
12                sum += (1 << i) * b;
13                System.out.print(sum + " ");
14            }
15            System.out.println();
16        }
17        sc.close();
18    }
19 }
20
```

Line: 20 Col: 1

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7.

HackerRank | Prepare | Java | Introduction | Java Datatypes | Exit Full Screen View

Problem

Java has 8 primitive data types: char, boolean, byte, short, int, long, float, and double. For this exercise, we'll work with the primitives used to hold integer values (byte, short, int, and long):

- A byte is an 8-bit signed integer.
- A short is a 16-bit signed integer.
- An int is a 32-bit signed integer.
- A long is a 64-bit signed integer.

Given an input integer, you must determine which primitive data types are capable of properly storing that input.

To get you started, a portion of the solution is provided for you in the editor.

Reference: <https://docs.oracle.com/javase/tutorial/java/nutsandbolts/datatypes.html>

Input Format

The first line contains an integer, T , denoting the number of test cases.

Each test case, T , is comprised of a single line with an integer, n , which can be arbitrarily large or small.

Output Format

For each input variable n and appropriate primitive **dataType**, you must determine if the given primitives are capable of storing it. If yes, then print:

```
n can be fitted in:
* dataType
```

Submissions

Leaderboard

Discussions

```
1 import java.util.Scanner;
2 public class Solution {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int t = sc.nextInt();
6         for (int i = 0; i < t; i++) {
7             try {
8                 long x = sc.nextLong();
9                 System.out.println(x + " can be fitted in:");
10                if (x >= Byte.MIN_VALUE && x <= Byte.MAX_VALUE)
11                    System.out.println(" byte");
12                if (x >= Short.MIN_VALUE && x <= Short.MAX_VALUE)
13                    System.out.println(" short");
14                if (x >= Integer.MIN_VALUE && x <= Integer.MAX_VALUE)
15                    System.out.println(" int");
16                System.out.println(" long");
17            } catch (Exception e) {
18                System.out.println(sc.next() + " can't be fitted anywhere.");
19            }
20        }
21    }
22    sc.close();
23 }
24
25
```

Line: 25 Col: 1

Upload Code as File | Test against custom input | Run Code | Submit Code

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8.

HackerRank | Prepare | Java | Introduction | Java End-of-file | Exit Full Screen View

Problem

"In computing, End Of File (commonly abbreviated EOF) is a condition in a computer operating system where no more data can be read from a data source." — (Wikipedia: End-of-file)

The challenge here is to read n lines of input until you reach EOF, then number and print all n lines of content.

Hint: Java's Scanner.hasNext() method is helpful for this problem.

Input Format

Read some unknown n lines of input from stdin(System.in) until you reach EOF; each line of input contains a non-empty String.

Output Format

For each line, print the line number, followed by a single space, and then the line content received as input.

Sample Input

```
Hello world
I am a file
Read me until end-of-file.
```

Sample Output

```
1 Hello world
```

Submissions

Leaderboard

Discussions

```
1 import java.util.Scanner;
2 public class Solution {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int lineNumber = 1;
6         while (sc.hasNextLine()) {
7             String line = sc.nextLine();
8             System.out.println(lineNumber + " " + line);
9             lineNumber++;
10        }
11    }
12    sc.close();
13 }
14
```

Line: 14 Col: 1

Upload Code as File | Test against custom input | Run Code | Submit Code

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9.

HackerRank

Prepare

Java

Introduction

Java Static Initializer Block

Exit Full Screen View

Problem

Static initialization blocks are executed when the class is loaded, and you can initialize static variables in those blocks.

It's time to test your knowledge of Static initialization blocks. You can read about it [here](#).

You are given a class `Solution` with a main method. Complete the given code so that it outputs the area of a parallelogram with breadth B and height H . You should read the variables from the standard input.

If $B \leq 0$ or $H \leq 0$, the output should be "java.lang.Exception: Breadth and height must be positive" without quotes.

Input Format

There are two lines of input. The first line contains B : the breadth of the parallelogram. The next line contains H : the height of the parallelogram.

Constraints

- $-100 \leq B \leq 100$
- $-100 \leq H \leq 100$

Output Format

If both values are greater than zero, then the main method must output the area of the parallelogram. Otherwise, print "java.lang.Exception: Breadth and height must be positive" without quotes.

Sample input 1

Submissions

Leaderboard

Discussions

Change Theme

Language

Java 15

```

1  import java.util.Scanner;
2  public class Solution {
3      static int B;
4      static int H;
5      static boolean flag = true;
6      static {
7          Scanner sc = new Scanner(System.in);
8          B = sc.nextInt();
9          H = sc.nextInt();
10         if (B <= 0 || H <= 0) {
11             flag = false;
12             System.out.println("java.lang.Exception: Breadth and height must be positive");
13         }
14         sc.close();
15     }
16     public static void main(String[] args) {
17         if (flag) {
18             int area = B * H;
19             System.out.println(area);
20         }
21     }
22 }
23

```

Line: 16 Col: 45

Upload Code as File

Test against custom input

Run Code

Submit Code

11

HackerRank

Prepare

Java

Introduction

Java Date and Time

Exit Full Screen View

Problem

The `Calendar` class is an abstract class that provides methods for converting between a specific instant in time and a set of calendar fields such as YEAR, MONTH, DAY_OF_MONTH, HOUR, and so on, and for manipulating the calendar fields, such as getting the date of the next week.

You are given a date. You just need to write the method, `getDay`, which returns the day on that date. To simplify your task, we have provided a portion of the code in the editor.

Example

month = 8
day = 14
year = 2017

The method should return *MONDAY* as the day on that date.

Submissions

Leaderboard

Discussions

Change Theme

Language

Java 15

```

1  import java.util.*;
2  import java.time.*;
3  public class Solution {
4      public static String findDay(int month, int day, int year) {
5          LocalDate date = LocalDate.of(year, month, day);
6          return date.getDayOfWeek().toString();
7      }
8      public static void main(String[] args) {
9          Scanner sc = new Scanner(System.in);
10         int month = sc.nextInt();
11         int day = sc.nextInt();
12         int year = sc.nextInt();
13         System.out.println(findDay(month, day, year));
14         sc.close();
15     }
16 }
17

```

Line: 17 Col: 1

Upload Code as File

Test against custom input

Run Code

Submit Code

10.

HackerRank | Prepare | Java | Introduction | Java Int to String | Exit Full Screen View

Problem

You are given an integer n , you have to convert it into a string.

Please complete the partially completed code in the editor. If your code successfully converts n into a string s the code will print "Good job". Otherwise it will print "Wrong answer".

n can range between -100 to 100 inclusive.

Sample Input 0

100

Sample Output 0

Good job

Language Java 15

```

1 import java.util.*;
2 public class Solution {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int n = sc.nextInt();
6         sc.close();
7         String s = Integer.toString(n);
8         if (n == Integer.parseInt(s)) {
9             System.out.println("Good job");
10        } else {
11            System.out.println("Wrong answer");
12        }
13    }
14 }
15

```

Line: 15 Col: 1

Upload Code as File Test against custom input Run Code Submit Code

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12

HackerRank | Prepare | Java | Strings | Java Strings Introduction | Exit Full Screen View

Problem

"A string is traditionally a sequence of characters, either as a literal constant or as some kind of variable." — Wikipedia: String (computer science)

This exercise is to test your understanding of Java Strings. A sample String declaration:

String myString = "Hello World!"

The elements of a String are called characters. The number of characters in a String is called the length, and it can be retrieved with the String.length() method.

Given two strings of lowercase English letters, A and B , perform the following operations:

- Sum the lengths of A and B .
- Determine if A is lexicographically larger than B (i.e.: does B come before A in the dictionary?).
- Capitalize the first letter in A and B and print them on a single line, separated by a space.

Input Format

The first line contains a string A . The second line contains another string B . The strings are comprised of only lowercase English letters.

Output Format

There are three lines of output:

For the first line, sum the lengths of A and B .

For the second line, write Yes if A is lexicographically greater than B otherwise print No

Language Java 15

```

1 import java.util.Scanner;
2 public class Solution {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         String A = sc.next();
6         String B = sc.next();
7         System.out.println(A.length() + B.length());
8         if (A.compareTo(B) > 0)
9             System.out.println("Yes");
10        else
11            System.out.println("No");
12        A = A.substring(0, 1).toUpperCase() + A.substring(1);
13        B = B.substring(0, 1).toUpperCase() + B.substring(1);
14        System.out.println(A + " " + B);
15        sc.close();
16    }
17 }
18

```

Line: 18 Col: 1

Upload Code as File Test against custom input Run Code Submit Code

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13.

14.

15.

HackerRank | Prepare | Java | **BigNumber** | Java Primality Test | Exit Full Screen View

Problem
A prime number is a natural number greater than 1 whose only positive divisors are 1 and itself. For example, the first six prime numbers are 2, 3, 5, 7, 11, and 13.

Given a large integer, n , use the Java BigInteger class' `isProbablePrime` method to determine and print whether it's prime or not prime.

Input Format
A single line containing an integer, n (the number to be checked).

Constraints
• n contains at most 100 digits.

Output Format
If n is a prime number, print `prime`; otherwise, print `not prime`.

Sample Input
13

Sample Output
prime

Explanation
The only positive divisors of 13 are 1 and 13, so we print `prime`.

Language: Java 15

```

1 import java.math.BigInteger;
2 import java.util.Scanner;
3 public class Solution {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         BigInteger n = sc.nextBigInteger();
7         if (n.isProbablePrime(100)) {
8             System.out.println("prime");
9         } else {
10            System.out.println("not prime");
11        }
12        sc.close();
13    }
14 }
15

```

Line: 15 Col: 1

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16.

HackerRank | Prepare | Java | **Data Structures** | Java Generics | Exit Full Screen View

Problem
Generic methods are a very efficient way to handle multiple datatypes using a single method. This problem will test your knowledge on Java Generic methods.

Let's say you have an integer array and a string array. You have to write a **single** method `printArray` that can print all the elements of both arrays. The method should be able to accept both integer arrays or string arrays.

You are given code in the editor. Complete the code so that it prints the following lines:

```

1
2
3
Hello
World

```

Do not use method overloading because your answer will not be accepted.

Language: Java 15

```

1 public class Solution {
2     static class Printer {
3         public <T> void printArray(T[] array) {
4             for (T element : array) {
5                 System.out.println(element);
6             }
7         }
8     }
9     public static void main(String[] args) {
10        Printer printer = new Printer();
11        Integer[] intArray = {1, 2, 3};
12        String[] stringArray = {"Hello", "World"};
13        printer.printArray(intArray);
14        printer.printArray(stringArray);
15    }
16 }
17

```

Line: 17 Col: 1

Upload Code as File | Test against custom input | Run Code | Submit Code

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17.

18.

19.

20.